

controllers used in financial applications, set-top boxes and the trusted platform module chips used in computers.

Over the next few years, we can expect to see the protocols used to initially configure and upgrade FPGAs and cSoCs improve to provide more features and improved security. New use models that help reduce costs will become available, especially where security demands are elevated, such as when board and system manufacturing is done by contract manufacturers, or in more sensitive applications such as systems used for national defence or homeland security. Field reprogramming will become more automated, simpler and more secure. In addition, supply chain issues such as counterfeit devices will be largely mitigated by appropriate countermeasures.

End users implementing data security applications will find new features aimed at making their designs more efficient. Implementing DPA countermeasures at the data security

application level will also become the norm over time. A nascent industry of specialised third-party IP providers is developing to produce high-quality, proven DPA-resistant soft cryptographic logic and firmware IP.

Some FPGAs and cSoCs will be developed with more highly evolved anti-tamper countermeasures, akin to smartcard microcontrollers. These devices will be self-contained single-chip security modules and highly flexible security processors that can be used where design and data security demands are highest. Coupled with board-level countermeasures, they will be able to satisfy the needs of all but the very highest security applications, without the need for developing an ASIC.

In a nutshell

The threat level for FPGAs and cSoCs will continue to increase as attacks get better. At the same time, more applications are demanding enhanced

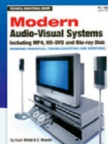
data security. For example, industrial controls and medical devices that in the past barely considered security are more and more considering it a prime design requirement. This trend is being accelerated by machine-to-machine applications and the 'Internet of Things.' Anti-tamper requirements in both defence and commercial applications are on the rise as everyone tries to stem the tide of IP theft and protect his company and country from the potentially disastrous consequences.

FPGAs and cSoCs are evolving to address current and anticipated device and system vulnerabilities. A new breed of devices that incorporate logic flexibility of a traditional FPGA, software flexibility of a microcontroller and security of a smartcard IC will be available within a few device generations. It's an exciting time for FPGA security! ●

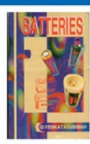
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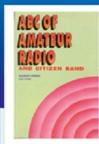
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